

Another Recipe App: A Design Case with Motive Disposition Theory

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ABSTRACT

As an interdisciplinary field, HCI studies are often built upon findings from other disciplines, such as the prevalent application of self-determination theory in HCI. Besides self-determination theory, motive disposition theory (MDT) is another prominent motivation theory that addresses human needs. However, the application of MDT is barely seen in HCI studies. MDT emphasizes individual differences in motives, differentiates implicit and explicit motives, and takes fear motives into account. We demonstrate a design case where MDT is applied to the design of a recipe app in order to motivate user engagement and deliver positive experiences. We then discuss our reflections from designing with MDT and identify concerns that could be addressed in the future. We hope that through our design case, the possibility and potential benefits of utilizing MDT in HCI design could be seen, and more discussions and studies with MDT in HCI could be sparked.

CCS CONCEPTS

• **Human-centered computing** → HCI design and evaluation methods; HCI theory, concepts and models; Interaction design theory, concepts and paradigms.

KEYWORDS

design, motive disposition theory, motivation, user experience

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1 INTRODUCTION

Psychological theories are commonly used in the field of HCI to explain behavior and inform design, particularly in areas such as behavior change, experience, and wellbeing. For instance, many studies used the theory of planned behavior [1] from psychology to

analyze behavior and as the theoretical framework to design behavioral interventions; notable examples include examining psychosocial determinants of pro-environmental behavior [5] and designing behavioral interventions to discourage students from using cars [6]. Positive psychology [47] serves as the foundation of the design framework of positive technology [38] and positive design [16]. Self-determination theory (SDT; [15]) is another popular psychological theory applied in many HCI studies, including the design guidelines for conversational agents [56] and motivational design patterns [26]. Yet, there is still a void in the application of motivation psychology in HCI.

Need theories in motivation psychology are one approach to explaining human behavior. Their central statement is that motives are expressed by human needs (e.g., physiological needs and psychological needs) [40]. Research has also shown that the fulfillment of psychological needs when interacting with technologies leads to a better user experience [21, 22, 53]. Hence, people would feel motivated and satisfied when engaging in activities that fulfill their psychological needs. For example, experience design [20] advocates for addressing universal psychological needs (e.g., security, competence, and popularity) to deliver good experiences. These needs are from psychology research [48] that draws upon several psychological theories, including SDT [53]. SDT is one theory about needs and motivation and arguably the most prevalent-used motivational theory in HCI studies [4]. However, there is another prominent theory regarding needs in motivation psychology but rarely mentioned in the field of HCI – the motive disposition theory (MDT; [28]).

Similar to SDT, MDT uses the concept of human needs to explain human motivation and personality [45, 49]. Unlike SDT, which considers the needs for competence, autonomy, and relatedness applying to all individuals in the same way, an important claim of MDT is that individuals are different, they have different motive dispositions. As a well-developed psychology theory, MDT has been examined in various settings [19, 24, 25, 46, 50, 55]. Nonetheless, in the field of HCI, only very few studies are involved with MDT (e.g., studies that use MDT to explain and predict gaming behavior by Poeller et al. [34, 37]). Poeller et al. [36] postulated that this is likely due to MDT's complexity and lack of digestible theory overview. To our knowledge, there is currently no design study based on MDT in the ACM digital library. As an important theory in motivation psychology, MDT could bring new insights into the field of HCI, especially regarding design. Poeller and Phillips [35] also suggested that the overuse of SDT in motivational research in HCI could be a pitfall since SDT is not a "theory of everything" (e.g., it does not

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take interindividual differences in need strength into consideration), and they advocated for including other motivation theories in HCI studies considering every theory has its own advantages to explain human behavior and to benefit HCI research. Therefore, we practiced using MDT to design digital artifacts that aim to motivate user engagement, and we showcase our use of MDT in the design of a recipe app in this paper.

2 MOTIVE DISPOSITION THEORY

Motivational research in psychology seeks the reasons for human behavior [23]. MDT treats motives as one of the determinants of action [28]. Motives are expressions of needs [40]; they energize and direct behavior to satisfy the needs, which means they influence which action options individuals see, choose, and persevere toward [28, 34]. A motive must first be activated by a corresponding incentive (environmental stimuli associated with satisfying this motive and then experiencing a rewarding feeling such as feelings of pride, strength, or interpersonal attachment) to influence behavior [10].

2.1 Implicit Motives vs. Explicit Motives

MDT differentiates between implicit and explicit motives. *Explicit motives* are goals individuals attribute to themselves and are consciously describable (e.g., the goal of reducing household energy use) [30]. They are affected by rational incentives (e.g., social expectations, demands, and external rewards), respond particularly to verbal cues (e.g., recognition and appreciation from others), and reflect an individual's self-image, beliefs, and values; thus, they particularly influence short-term behavior based on consciously elaborated decisions [7, 30, 43, 45]. *Implicit motives*, on the other hand, are affectively charged preferences for incentives inherent in the activity or task itself (e.g., difficulty and novelty in certain tasks) [10]. They are considered inaccessible to introspection and are sensitive to nonverbal signals such as facial expressions [10, 30, 43]. Unlike explicit motives, implicit motives are considered relatively stable; thus, they are considered to have effects on spontaneous and long-term behavior [7, 30].

2.2 The “Big Three”

Instead of considering that needs are universal between individuals, MDT posits that individuals are different in motives (hence different people have different motive dispositions and have behavioral differences across time and situations). There are three implicit motives (so-called the “big three”) that have the most advanced research today: achievement motive, affiliation motive, and power motive. Individuals can be high or low in each motive.

Achievement motive (or need for achievement) is a preference for activities that give rewarding experiences and involve competition with a standard of excellence [3, 11, 32]. For example, to work on challenging mathematical problems. Achievement-motivated individuals are capable of deriving satisfaction from the autonomous mastery of challenging tasks [29, 42, 43]. *Affiliation motive* (or need for affiliation) is the desire to establish, maintain, or restore positive emotional relationships [8, 52]. For example, to make new friends at social events. Affiliation-motivated individuals get satisfaction from close, harmonious contact with others [28, 52]. *Power motive* (or need for power) is the need to influence, control, or impress others

and to get recognition for these types of behavior [18, 28]. For example, to win a verbal argument or to lead a team. Power-motivated individuals obtain pleasure from impacting or dominating others [18, 28].

2.3 Fear vs. Hope

In MDT, each implicit motive has two components — fear and hope. Accordingly, in the achievement motive, there are *hope for success* and *fear of failure*; in the affiliation motive, there are *hope for affiliation* and *fear of rejection*; in the power motive, there are *hope for power* and *fear of weakness*. Individuals who are fear-motivated also have hope to satisfy their motives, but their fear of failure, rejection, and weakness make them act differently than individuals without the fear [45]. For example, people who are high in hope for success might prefer medium-difficulty tasks for the higher possibility of getting the rewarding feeling of success; in contrast, people who are high in fear of failure would avoid medium-difficulty tasks and tend to choose either the easy tasks to ensure they would not fail or the very difficult tasks since not completing them is not an indication of inadequate ability [11].

3 ANOTHER RECIPE APP

3.1 Background

The concept of the recipe app was initially developed as part of a master's thesis project on designing to promote edible insects in the West. Edible insects are good alternatives to meat products due to their nutritious and environmentally friendly qualities [14, 31, 39], but the consumption of edible insects in the West is very low [2, 13, 17]. According to the theory of planned behavior [1], whether a Westerner would consume edible insects is related to this person's attitude towards consuming edible insects (could be improved by informing people about the benefits of consuming edible insects), this person's subjective norm of consuming edible insects (could be changed by knowing other people are already consuming edible insects), and this person's perceived difficulty in consuming edible insects (could be lightened by giving instructions on how to access to insect-based food).

We decided on the idea of a recipe app after a brainstorming session. First, since edible insects are food, a recipe app would be a good place to introduce edible insects and inform people of their benefits. Second, well-made insect-based food recipes presented nicely in a recipe app would further change people's attitude towards consuming edible insects to the positive side. Third, a recipe app with the motif of sustainable food could attract people who care about the environment, and this type of person is more likely to gravitate towards trying insect-based recipes due to the environmental benefit of consuming edible insects. As a result, this group of people could be the first adopter of edible insects and gradually change the norm. Fourth, recipe instructions in a recipe app teach people how to cook insect-based food, which could reduce the perceived difficulty in consuming edible insects. Altogether, we considered a recipe app the best means to promote edible insects in the West. More importantly, we applied MDT in designing the app in order that people would feel motivated to engage with it.

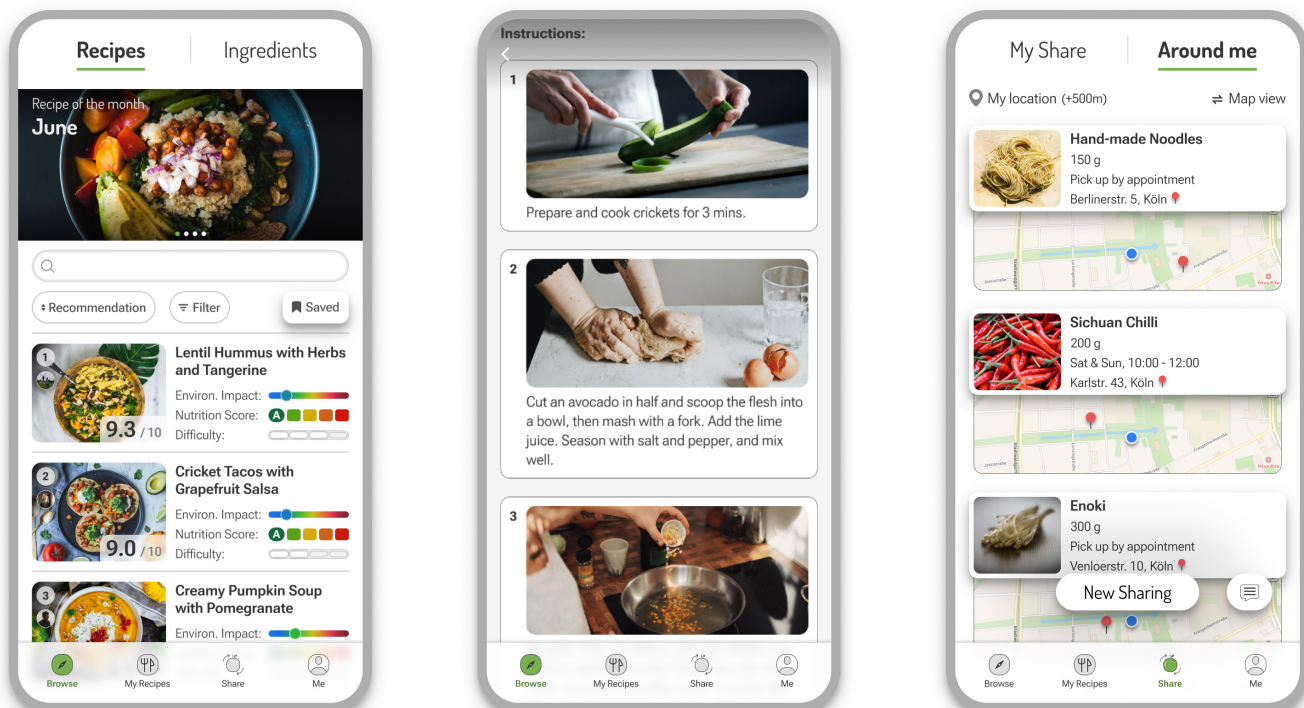


Figure 1: On the left: the recipe browsing page showing that each recipe has its rating and difficulty level. In the middle: detailed cooking instructions showing inside a recipe. On the right: the ingredient-sharing feature that supports users in sharing and requesting surplus ingredients.¹

3.2 Applying MDT in Design

According to MDT, implicit motives (expressions of needs) can be activated by corresponding incentives, which then energize and direct behavior to fulfill the underlying needs. Therefore, designers could design artifacts that include activities and tasks containing these incentives to activate people's implicit motives. When people are interacting with the design artifact, their implicit motives could be activated and they would engage in the activities and tasks to satisfy their motives. Also, the artifact could also be designed to facilitate people on the activities and tasks. When people's activated implicit motives get satisfied, they experience rewarding feelings and enjoy engaging with the design artifact. While the artifact itself might not directly motivate people, the activities and tasks within the design that contain incentives could. The artifact also does not directly provide people with rewarding feelings, but it facilitates them in performing the activities and tasks which are the real causes of their satisfaction. Taken together, an MDT-based design could motivate people to engage and convey positive experiences.

To design with MDT, three aspects need to be considered. First, all three implicit motives need to be considered. Since MDT assumes that individuals are different in motives, a design that only

considers one or two implicit motives would not motivate all types of people. For example, if a design artifact only has features for achievement motive, users who are low in achievement motive but high in affiliation or power motive might be less motivated to use. Second, since people can be fear-motivated, an MDT-based design also needs to consider the risks of experiencing failure, rejection, and weakness. Lowering these risks could encourage fear-motivated people to engage and satisfy their motives. Last but not least, implicit affiliation and power motives are social motives; therefore, an MDT-based design should take the interactions between people into consideration.

3.3 The MDT-based Recipe App

Our design of the recipe app has three main functionalities: 1. browsing recipes, 2. creating recipes, and 3. sharing ingredients.

3.3.1 Features for Implicit Achievement Motive. The recipe app shows food recipes and detailed instructions (as shown in Figure 1 left and middle), and users' implicit achievement motive could be satisfied by successfully cooking a nice meal. There is a difficulty level assigned to each recipe so that fear-motivated users can choose easy recipes to ensure success. Users could also gain the feeling of success by creating quality recipes and receiving high ratings from other users. Even though the implicit achievement motive is about mastering challenging tasks and competing with a standard

¹Sources of photo in the prototype: on the left from top to bottom, photos by Sonny Mauricio, Jo Sonn, Shannon Nickerson, and Ruth Georgiev on Unsplash; in the middle from top to bottom, photos by CA Creative, Alex Lvr, and Conscious Design on Unsplash; on the right from top to bottom, photos by Tommy Johns, Elle Hughes, and Nathalie Jolie on Unsplash.

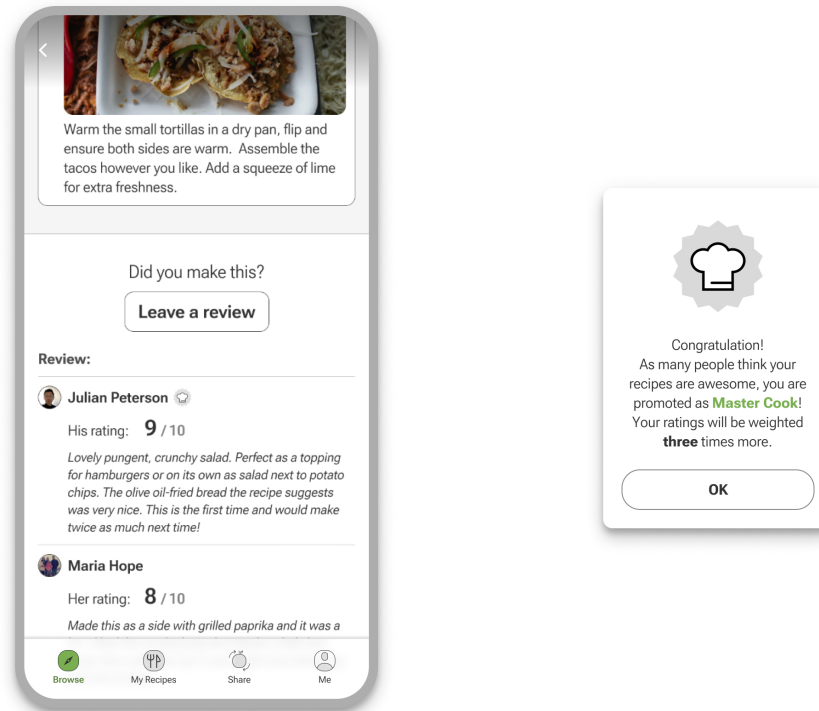


Figure 2: On the left: users rating and reviews inside a recipe. On the right: a pop-up notification regarding *Master Cook*.²

of excellence rather than other users, ratings could give users a reference and help them set their standard of excellence.

3.3.2 Features for Implicit Affiliation Motive. The recipe app also supports users in sharing surplus ingredients with other users nearby (as shown in Figure 1 right), and this functionality could lead to the satisfaction of users' implicit affiliation motive. Sharing food requires contact with other people; therefore, it creates opportunities to establish relationships with others. For example, when picking up a shared ingredient, users could chat about their preferred way to cook this ingredient. As to users who have fear of being rejected, they could choose to only share their surplus ingredients and not ask for what other users are sharing.

3.3.3 Features for Implicit Power Motive. The recipe rating functionality in the app (as shown in Figure 2 left) is designed to satisfy users' implicit power motive. Users could show their influence over others by giving ratings to recipes that are created by other users. For users who fear that their creations could be criticized by other users, they could set their recipes to private, in order that no rating or review can be given. However, if users have more hope for power, they have the chance to be promoted as *Master Cook* in the app (as shown in Figure 2 right). A *Master Cook* has a bigger voice in a recipe's rating. A rating from a *Master Cook* weighs three times more than a regular user's when calculating the overall score of a recipe; therefore, a *Master Cook* could have more impact on other

users. To ensure that users with more power are qualified, users must have three high-rating recipes of their own to earn the title *Master Cook*.

In summary, the recipe app was designed to consider all three implicit motives, as well as the hope and fear components within each motive. As a result, the app has the potential to motivate people to use it and bring positive experiences to them, regardless of which implicit motives they have.

4 DISCUSSION

Although the design of the recipe app was not evaluated, we have several reflections from designing with MDT.

4.1 Designing for Fear

The main challenge we faced when we were designing with MDT is to design for people with fear motives. It is hard to motivate this kind of people because when they face the chance to satisfy their motives, what they are seeing is mainly the risk of having the opposite results. Therefore, we tried to design functionalities that do not bring high risks of experiencing failure, rejection, and weakness. The hope-motivated people, on the other hand, are relatively easy to design for; they do not worry about the risk of experiencing the opposite of what they want, so they are free to do what would satisfy their motives.

How to design for fear-motivated people could be interesting to research in the future, especially considering that there are two

²Source of photo in the prototype: on the left at the top, photo by angela pham on Unsplash.

possible kinds of fear reactions (i.e., active avoidance: actively behaving to avoid the potential negative outcome; passive avoidance: simply doing nothing to avoid the potential negative outcome [44].) Furthermore, since fear-motivated people may refrain from doing the activities and tasks that can satisfy their implicit motives, they get less satisfaction, which could impair their psychological wellbeing. However, motivating individuals through their fear may raise ethical concerns, and the experience can suffer from it.

4.2 Two Faces of Power

The power motive is the need to influence other people, and could actually be expressed in two different forms of behavior — personalized (antisocial) and socialized (prosocial) [27, 45, 54]. Individuals who express power in a personalized way develop their influence without caring much for others' interests, and they usually do not get painted in a positive way [12]. On the other hand, power can also be expressed in a socialized way that is beneficial to others, such as via helping and teaching others. The design of the recipe app did not take the two faces of power into consideration, but one future work could be designing functionalities that are related only to socialized power.

4.3 Certainty of Choice

Even though the design of the recipe app considered individual differences in implicit motives, there is no mechanism to ensure that users will use the functionalities that match their implicit motives. This could be due to the influence of explicit motives, which are accessible to introspection, unlike implicit motives. As a result, people might behave in a manner consistent with their explicit motives rather than their implicit motives. For example, a user who is explicitly achievement-motivated but actually has a low implicit achievement motive and high implicit affiliation motive might often aim for creating exceptional recipes to receive high ratings rather than participating in ingredient-sharing activities, which could help them build connections with others. This can be problematic since the user's low implicit achievement motive might result in a lack of spontaneous inclination to work hard. In the end, this user might experience neither achievement nor affiliation through the app and would not feel motivated to engage with the app. The discordance between explicit motive and implicit motive creates motive incongruence, which is commonly considered to be undesirable and can cause a variety of negative outcomes including decrements in emotional wellbeing and life satisfaction [9, 51].

However, as implicit motives are sensitive to nonverbal incentives, another future work could be regarding how to use this quality of implicit motives to increase the certainty that the choices of activities and tasks from users match their implicit motives.

4.4 Need State and Quality of the Incentive

Another potential issue is related to the need state. As motivation is dynamic, individuals' need states could be different at different times (e.g., before and after a need just being fulfilled), and consequently affects the interaction with the app (incentives can be more attractive when the need state is high). Need state would also be activated to different levels for different individuals with the same incentive because the nature or quality of the incentive itself

also matters (e.g., some people might be naturally uninterested in cooking) [44]. This implies that it might be challenging to have a one-size-fits-all design.

4.5 Evaluation Strategy

The evaluation strategy was not yet fully explored, but we have already anticipated a couple of challenges. Since the purpose of designing with MDT is to motivate people to engage with the design, the evaluation should focus on whether the MDT-based design effectively motivates people to use it. To accomplish this, a functional prototype should be created and tested by actual users. However, in our case, since the recipe app is community-based, a challenge would be how to efficiently create a functional prototype that involves numerous users. Another potential challenge concerns the time needed for evaluation. Since implicit motives predict spontaneous and long-term behavior, to better determine the effectiveness of the MDT-based design on people, longer-term testing of the design might be necessary.

Apart from the user testing, users' implicit and explicit motives need to be measured, such as the Picture Story Exercise [33] for implicit motives and the Unified Motive Scales [41] for explicit motives. Additionally, an interview needs to be conducted at the end of the testing to explore how users used the artifact and their experience of using it. The reasons for conducting these measurements and interviews are twofold. On the one hand, if users felt motivated to use certain functionality, we need to confirm if the functionality they liked is the one that is designed for their implicit motives (e.g., if a power-motivated user enjoys using the feature that is designed for power motive); if so, this may imply that the MDT-based design is effective. On the other hand, if users did not feel motivated to use certain functionality, we need to investigate if it is because of the influence of their explicit motives, the nature of the incentive in the functionality, or other reasons.

4.6 Implications for Behavior Change Design

One of the ways digital artifacts are being used in HCI is as the medium to facilitate behavior change, such as our recipe app for promoting edible insects. The recipe app informs people of the benefits of edible insects and helps make insect-based food accessible; therefore, it supports the behavior change from not consuming insect-based food to consuming it. However, this envisioning has a precondition: people have to engage with the recipe app. This implies that there is a gap between the designed artifact and the intended impact of the artifact. That is why we applied motivation psychology in the design of the recipe app to motivate user engagement. Hence, as a prominent theory in motivation psychology, MDT is worth being explored in the context of behavior change in HCI.

5 CONCLUSION

In this paper, we first pointed out the fact that MDT as an important motivation theory has not been applied in HCI design. We then demonstrated our use of MDT in the design of a recipe app, in which we considered all three motives and their hope and fear components. We discussed our reflections on the design, in which we believed that there were several concerns that we missed but

could be addressed in the future when designing with MDT. We hope our design case showed the possibility and potential benefits of designing with MDT and more studies incorporating MDT could be seen in HCI.

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